Nonlinear functions — Part 2 answer key

#	ID	Answer	Difficulty
2.1	a598add1-606e-453f-8b43-02083a30f5c7	C	Medium
2.2	969792b2-4ab7-4fe4-8d9e-e1a709101015	A	Medium
2.3	025702-DC	D	Medium
2.4	94ede15a-23c9-4c00-bcde-09c2bb206b79	C	Medium
2.5	029384-DC	C	Medium
2.6	6951ac71-d64f-4a32-b84a-46f30f206692	410	Medium
2.7	407267f8-d51c-4b0c-bd4e-fec1fe7b71c8	.5, 1/2	Medium
2.8	d900a2e7-10e1-484e-a11a-0c2fe3bbb04f	B	Medium
2.9	30c5bb08-3642-4126-a066-371cb194f2d1	B	Medium
2.10	025038-DC	D	Medium
2.11	375d70d6-9387-4fb7-a51e-674183bfce7c	B	Medium
2.12	885439e9-2f0d-44e8-a3af-605c5924be71	A	Medium
2.13	f41f1cb2-304e-4865-a112-f9de1e665a43	C	Medium
2.14	ceec8040-0ddb-4726-b897-3baa00b904d2	B	Medium
2.15	5186bb0a-09e9-4581-aae8-e440d02355fe	B	Medium
2.16	025805-DC	D	Medium
2.17	585130f5-a566-4db1-900e-4f8d08cd67e6	B	Medium
2.18	b76e3ad3-f70a-40d5-b897-a3b5e368da81	C	Medium
2.19	05eb7086-86be-4c43-bb55-070f148ab34a	D	Medium
2.20	e10027a7-cac8-4ca2-855a-ca3f00b55687	B	Medium
2.21	3f8c2f2d-abee-4738-aab3-11a77be98e29	B	Medium
2.22	98b110ad-3cb9-4925-831e-09572e226fda	D	Medium

2.23	09291-DC	C	Medium
2.24	664fb796-9a6b-4891-a961-52cdf5f5b989	D	Medium
2.25	ca334282-00d9-4e64-9f61-edc6bc5d7646	C	Medium
2.26	ce714e5d-1fb7-44e8-bfbc-55e6659cac72	C	Medium
2.27	f8a92cc0-95bf-4c16-8e6e-b31fa01f8b52	B	Medium
2.28	070617-DC	?	Medium
2.29	a8304c28-11ca-4f4b-b7bf-4379eab3e67e	B	Medium
2.30	4c9d9c55-af83-41a1-beaf-3cf5e261efd5	B	Medium
2.31	1b3396a5-95c8-4241-a154-8fc5c60e081b	D	Medium
2.32	61c05d04-1600-452b-9723-85d1636b8f5a	B	Medium
2.33	028933-DC	B	Medium
2.34	9025546d-7856-480c-9be8-6cbd532f45ee	D	Medium
2.35	46f2fda7-d91a-4059-9f4c-9f57b65cf702	B	Medium
2.36	853afe28-d4da-47b1-abf4-817c4a543239	B	Medium
2.37	8472776c-becb-4a82-bf1b-aa29eb25da77	A	Medium
2.38	85dc4466-29e1-4dd1-8c9b-cf1c1f0d641f	A	Medium
2.39	d73707f3-c5fe-430b-8799-06ef95532995	D	Medium
2.40	abfd9618-c65d-4fd8-9f89-254e3dfa9529	A	Medium
2.41	029475-DC	A	Medium
2.42	ccdb450a-31b3-42ec-9532-4859dd3972c3	A	Medium
2.43	d0812cf6-f4e9-4df1-9620-878d9a37d592	18	Medium
2.44	b43f007e-8a7c-48b8-8a0a-eeb0f65e3faf	D	Medium
2.45	0ef7164f-e18c-4aa3-999c-ece225b93166	C	Medium
2.46	45800f0b-d83c-4237-a7a2-1f84edd039a1	B	Medium
2.47	59addf61-9f46-4c0d-9dce-ebce1fa135b7	11/4, 2.75	Medium
2.48	d7821fa6-7c23-42a3-8896-9b2a32bffb92	C	Medium

2.49	eb535e3d-3c71-4caf-883c-c6a73134317d	A	Medium
2.50	016491-DC	A	Medium
2.51	027803-DC	D	Medium
2.52	ecc55958-e485-4d44-84c5-3a70884d1fb5	B	Medium
2.53	433ee68c-fedc-4334-84d7-4032b35bf0e2	D	Medium
2.54	017018-DC	D	Medium
2.55	20b0ac3b-55d7-436b-be64-cb9086cccb62	68	Medium
2.56	044c0858-a7b2-4913-8535-7de0c2051f19	A	Medium
2.57	b026eab4-9126-4926-ba7f-aeb53aa41e01	17	Medium
2.58	dbd2580c-f8fe-48de-b7f0-838c5fe91f7c	B	Medium
2.59	ff8903c3-0d12-4c8b-8aef-3372dfbd9793	B	Medium
2.60	029350-DC	C	Medium
2.61	025772-DC	?	Medium
2.62	9accecc3-e743-4bfa-9e4f-235283d99abd	11875	Medium
2.63	8fbdc9d2-8c5f-4da3-98db-20599f2fff76	B	Medium
2.64	028940-DC	B	Medium
2.65	045551-DC	A	Medium
2.66	021775-DC	A	Medium
2.67	fdf648c1-f590-4c15-b580-2bbbdeb7df2f	B	Medium
2.68	7888cfcc-ba3d-422a-be02-86652b46cb9d	D	Medium
2.69	2a0b1f88-5e2b-441c-84cb-141f8592d02a	D	Medium
2.70	f9da1b3c-c8a6-40ae-b912-90cb932f1c07	D	Medium
2.71	ee6062be-ade1-48a3-b556-2cca3b7b45e3	B	Hard
2.72	7fc3b869-8f5c-4c0b-a28b-acbe8ea76ddf	B	Hard
2.73	04172-DC	D	Hard
2.74	baa55cdd-6e64-416a-8f18-d484e43741e8	D	Hard

2.75	fb591f40-b171-470c-ab62-48c9c0ee004b	D	Hard
2.76	2da8e72d-a24c-49d7-a9ed-5f47b4fecf08	168	Hard
2.77	19eed232-6ab6-4e1d-8586-46e6d6357a09	D	Hard
2.78	e6248b96-6cdd-48a4-a20b-dc4b62d59d56	C	Hard
2.79	72f977c3-5379-42b1-947f-d9fc4dbc27fb	A	Hard
2.80	35b45e44-2494-439f-a174-5c714299c55c	-324	Hard
2.81	59249867-1d27-418b-aac8-59cddb26069b	-7	Hard
2.82	65743096-f9d1-4ab0-9a10-3acd3ea46114	D	Hard
2.83	5af7e994-3850-4c5c-987d-1213c4f6baf8	C	Hard
2.84	c1e6cb10-5989-4a1e-a246-78972ed7bf5a	D	Hard
2.85	f78a967a-96bf-469b-bc93-679f3d5dac92	D	Hard
2.86	1010a838-bf8c-4e58-8c76-a23fa7c4ee2b	25/4, 6.25	Hard

Rationales

▼ 2.1 — C

Choice C is correct. It's given that the value of a certain fishing boat was $\$ 24{,}000$ when it was purchased. It's also given that each year, the value of the fishing boat decreased by 20% of the previous year's value. It follows that this situation can be represented by the equation $v = a1 - \frac{r}{100}^{y}$, where v is the value of the boat, in dollars, y years since it was purchased, a is the value of the boat, in dollars, when it was purchased, and r is the percent by which the value of the boat decreases each year from the previous year's value. Substituting $24{,}000$ for a and 20 for r in this equation yields $v = 24{,}0001 - \frac{20}{100}^{y}$, or $v = 24{,}0000.8^{y}$. Therefore, $v = 24{,}0000.8^{y}$ represents this situation.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This equation represents a situation where each year the value of the boat increased, not decreased, by 20% of the previous year's value.

▼ 2.2 — A

Choice A is correct. The function f gives the area of the rectangle, in ft^2, if its width is w ft. Since the value of $f14$ is the value of fw if $w = 14$, it follows that $f14 = 1{,}176$ means that fw is $1{,}176$ if $w = 14$. In the given context, this means that if the width of the rectangle is 14 ft, then the area of the rectangle is $1{,}176$ ft^2.

Choice B is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

Choice D is incorrect and may result from interpreting fw as the width, in ft, of the rectangle if its area is w ft^2, rather than as the area, in ft^2, of the rectangle if its width is w ft.

▼ 2.3 — D

Choice D is correct. The function N is exponential, so it follows that $N(d)$ changes by a fixed percentage for each increase in d by 1. Since d is measured in hundreds of meters, it also follows that the number of brachiopod species changes by a fixed percentage for each increase in ocean depth by 100 meters. Since the base of the exponent in the model is 0.90, which is less than 1, the number of brachiopod species decreases as the ocean depth increases. Specifically, the number of brachiopod species at a depth of $d+100$ meters is 90% of the number of brachiopod species at a depth of d meters. This means that for each increase in ocean depth by 100 meters, the number of brachiopod species decreases by 10%.

Choices A and C are incorrect. These describe situations where the number of brachiopod species are decreasing linearly rather than exponentially. Choice B is incorrect and results from interpreting the decrease in the number of brachiopod species as 10% for every 1-meter increase in ocean depth rather than for every 100-meter increase in ocean depth.

▼ 2.4 — C

Choice C is correct. It's given that $y = 7{,}400(0.87)^x$ estimates the value, in dollars, of a certain piece of equipment x years after it was purchased. The estimated value of the piece of equipment at the time it was purchased corresponds to the value of y when $x = 0$. Substituting 0 for x in the equation $y = 7{,}400(0.87)^x$ yields $y = 7{,}400(0.87)^0$, which is

equivalent to $y = 7{,}400(1)$, or $y = 7{,}400$. Therefore, the estimated value, in dollars, of the piece of equipment at the time it was purchased was $7{,}400$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This was the estimated value, in dollars, of the piece of equipment 1 year after it was purchased.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 2.5 — C

Choice C is correct. Substituting -1 for x in the given function f gives $f(-1) = (-1)^3 + 3(-1)^2 - 6(-1) - 1$, which simplifies to $f(-1) = -1 + 3(1) - 6(-1) - 1$. This further simplifies to $f(-1) = -1 + 3 + 6 - 1$, or $f(-1) = 7$.

Choice A is incorrect and may result from correctly substituting -1 for x in the function but incorrectly simplifying the resulting expression to $f(-1) = -1 - 3 - 6 - 1$, or -11. Choice B is incorrect and may result from arithmetic errors. Choice D is incorrect and may result from correctly substituting -1 for x in the function but incorrectly simplifying the expression to $f(-1) = 1 + 3 + 6 + 1$, or 11.

▼ 2.6 — 410

The correct answer is 410. It's given that t minutes after an initial observation, the number of bacteria in a population is $60{,}000 2^{\frac{t}{410}}$. This expression consists of the initial number of bacteria, $60{,}000$, multiplied by the expression $2^{\frac{t}{410}}$. The time it takes for the number of bacteria to double is the increase in the value of t that causes the expression $2^{\frac{t}{410}}$ to double. Since the base of the expression $2^{\frac{t}{410}}$ is 2, the expression $2^{\frac{t}{410}}$ will double when the exponent increases by 1. Since the exponent of the expression $2^{\frac{t}{410}}$ is $\frac{t}{410}$, the exponent will increase by 1 when t increases by 410. Therefore the time, in minutes, it takes for the number of bacteria in the population to double is 410.

▼ 2.7 — .5, 1/2

The correct answer is $\frac{1}{2}$. The value of $h2$ is the value of hx when $x = 2$. Substituting 2 for x in the given equation yields $h2 = \frac{8}{5^{2+6}}$, which is equivalent to $h2 = \frac{8}{16}$, or $h2 = \frac{1}{2}$. Therefore, the

value of $h2$ is $\frac{1}{2}$. Note that 1/2 and .5 are examples of ways to enter a correct answer.

▼ 2.8 — B

Choice B is correct. It's given that the graph models the number of active projects a company was working on x months after the end of November 2012. Therefore, the value of x that corresponds to the end of November 2012 is 0. The point at which $x = 0$ is the y-intercept of the graph. It follows that the y-intercept of the graph shown is the point $0, 5$. Therefore, according to the model, the predicted number of active projects the company was working on at the end of November 2012 is 5.

Choice A is incorrect. This is the value of x that corresponds to the end of November 2012, not the predicted number of active projects the company was working on at the end of November 2012.

Choice C is incorrect. This is the predicted number of active projects the company was working on 2 months after the end of November 2012.

Choice D is incorrect. This is the predicted number of active projects the company was working on 4 months after the end of November 2012.

▼ 2.9 — B

Choice B is correct. Each of the given choices is an equation of the form $y = -\frac{1}{10}x(x - a)^m(x + b)^n$, where a, b, m, and n are positive constants. In the xy-plane, the graph of an equation of this form has x-intercepts at $x = 0$, $x = a$, and $x = -b$. The graph shown has x-intercepts at $x = 0$, $x = 4$, and $x = -5$. Therefore, $a = 4$ and $b = 5$. Of the given choices, only choices A and B have $a = 4$ and $b = 5$. For an equation in the form $y = -\frac{1}{10}x(x - a)^m(x + b)^n$, if all values of x that are less than $-b$ or greater than a correspond to negative y-values, then the sum of all the exponents of the factors on the right-hand side of the equation is even. In the graph shown, all values of x less than -5 or greater than 4 correspond to negative y-values. Therefore, the sum of all the exponents of the factors on the right-hand side of the equation $y = -\frac{1}{10}x(x - 4)^m(x + 5)^n$ must be even. For choice A, the sum of these exponents is $1 + 1 + 1$, or 3, which is odd. For choice B, the sum of these exponents is $1 + 1 + 2$, or 4, which is even. Therefore, $y = -\frac{1}{10}x(x - 4)(x + 5)^2$ could be the equation of the graph shown.

Choice A is incorrect. For the graph of this equation, all values of x less than -5 correspond to positive, not negative, y-values.

Choice C is incorrect. The graph of this equation has x-intercepts at $x = -4$, $x = 0$, and $x = 5$, rather than x-intercepts at $x = -5$, $x = 0$, and $x = 4$.

Choice D is incorrect. The graph of this equation has x-intercepts at $x = -4$, $x = 0$, and $x = 5$, rather than x-intercepts at $x = -5$, $x = 0$, and $x = 4$.

▼ 2.10 — D

Choice D is correct. Because the change in the number of students decreases by the same percentage each year, the relationship between the number of students and the number of years can be modeled with a decreasing exponential function in the form $f(x) = a(1 - r)^x$, where $f(x)$ is the number of students, a is the number of students in 2015, r is the rate of decrease each year, and x is the number of years since 2015. It's given that 3,000 students were enrolled in 2015 and that the rate of decrease is predicted to be 2%, or 0.02. Substituting these values into the decreasing exponential function yields $f(x) = 3{,}000(1 - 0.02)^x$, which is equivalent to $f(x) = 3{,}000(0.98)^x$.

Choices A, B, and C are incorrect and may result from conceptual errors when translating the given information into a decreasing exponential function.

▼ 2.11 — B

Choice B is correct. For a quadratic function defined by an equation of the form $gx = ax - h^2 + k$, where a, h, and k are constants and $a > 0$, the minimum value of the function is k. In the given function, $a = 1$, $h = 0$, and $k = 55$. Therefore, the minimum value of the given function is 55.

Choice A is incorrect. This is the value of x for which the given function reaches its minimum value, not the minimum value of the function.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 2.12 — A

Choice A is correct. For an exponential function of the form $Ct = a(b)^t$, where a and b are constants, the initial value of the function, or the value of the function when $t = 0$, is a, and

the value of the function changes by a factor of b each time t increases by 1. It follows that in the given function, the initial value is 6 and the value of the function changes by a factor of 1.5 each time t increases by 1. It's given that the function C gives the estimated number of clients at a certain business and t is the number of weeks since the business opened. Therefore, the best interpretation of the number 6 in this context is the estimated number of clients when $t = 0$, or when the business opened.

Choice B is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

Choice D is incorrect and may result from conceptual errors.

▼ 2.13 — C

Choice C is correct. In the given equation, x represents the number of days after a new product launched, where $0 \leq x \leq 20$, and y represents the estimated stock price, in dollars, for a certain company. Therefore, the best interpretation of $x, y = 1, 83$ in this context is that 1 day after the new product launched, the company's estimated stock price is $\$ 83$.

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

Choice D is incorrect and may result from conceptual errors.

▼ 2.14 — B

Choice B is correct. Each of the given choices is an equation of the form $fx = ab^{x-k}$, where a, b, and k are constants. For an equation of this form, the coefficient, a, is equal to the value of the function when the exponent is equal to 0, or when $x = k$. It follows that in the equation $fx = 331.5^x$, the coefficient, 33, is equal to the value of $f0$. Substituting 0 for x in this equation yields $f0 = 331.5^0$, which is equivalent to $f0 = 331$, or $f0 = 33$. Thus, the value of c is 33 and the equation $fx = 331.5^x$ shows the value of c as the coefficient.

Choice A is incorrect. This equation shows the value of $f\text{-}1$, not $f0$, as the coefficient.

Choice C is incorrect. This equation shows the value of $f1$, not $f0$, as the coefficient.

Choice D is incorrect. This equation shows the value of $f2$, not $f0$, as the coefficient.

▼ 2.15 — B

Choice B is correct. It's given that $g(x) = 10^{5x-8}$. Since $10{,}000$ can be written as 10^4, substituting 10^4 for $g(x)$ in this equation yields $10^4 = 10^{5x-8}$. Because the bases on the left-hand side and right-hand side of this equation are equal, it follows that the exponents must be equal. Therefore, $4 = 5x - 8$. Adding 8 to both sides of this equation yields $12 = 5x$. Dividing both sides of this equation by 5 yields $\frac{12}{5} = x$. Therefore, the value of x when $g(x)$ is equal to $10{,}000$ is $\frac{12}{5}$.

Choice A is incorrect. This is the value of x when $g(x)$ is equal to $1{,}000$, not $10{,}000$.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the value of $5x - 8$, not x, when $g(x)$ is equal to $10{,}000$.

▼ 2.16 — D

Choice D is correct. Since t represents the number of years after 2010, the estimated data traffic, in terabytes, in 2010 can be calculated using the given equation when $t = 0$. Substituting 0 for t in the given equation yields $D = 5{,}640(1.9)^0$, or $5{,}640(1) = 5{,}640$. Thus, 5,640 represents the estimated data traffic, in terabytes, in 2010.

Choice A is incorrect. Since the equation is exponential, the amount of increase of data traffic each year isn't constant. Choice B is incorrect. According to the equation, the percent increase in data traffic each year is 90%. Choice C is incorrect. The estimated data traffic, in terabytes, for the year that is t years after 2010 is represented by D, not the number 5,640.

▼ 2.17 — B

Choice B is correct. It's given that the graph of $y = g(x)$ is produced by translating the graph of $y = f(x)$ 3 units down and 4 units to the right in the xy-plane. Therefore, function g can be defined by an equation in the form $g(x) = f(x - 4) - 3$. Function f is defined by the equation $f(x) = \frac{a - 19}{x} + 5$, where a is a constant. Substituting $x - 4$ for x in the equation $f(x) = \frac{a - 19}{x} + 5$ yields $f(x - 4) = \frac{a - 19}{x - 4} + 5$. Substituting $\frac{a - 19}{x - 4} + 5$ for $f(x - 4)$ in the equation $g(x) = f(x - 4) - 3$ yields $g(x) = \frac{a - 19}{x - 4} + 5 - 3$, or $g(x) = \frac{a - 19}{x - 4} + 2$. Therefore, the equation that defines function g is $g(x) = \frac{a - 19}{x - 4} + 2$.

Choice A is incorrect. This equation defines a function whose graph is produced by translating the graph of $y = fx$ 3 units down and 4 units to the left, not 3 units down and 4 units to the right.

Choice C is incorrect. This equation defines a function whose graph is produced by translating the graph of $y = fx$ 4 units to the left, not 3 units down and 4 units to the right.

Choice D is incorrect. This equation defines a function whose graph is produced by translating the graph of $y = fx$ 4 units to the right, not 3 units down and 4 units to the right.

▼ 2.18 — C

Choice C is correct. Since the value of qx decreases by a fixed percentage, 45%, for every increase in the value of x by 1, the function q is a decreasing exponential function. A decreasing exponential function can be written in the form $qx = a1 - \frac{p}{100}^x$, where a is the value of $q0$ and the value of qx decreases by p% for every increase in the value of x by 1. If $q0 = 14$, then $a = 14$. Since the value of qx decreases by 45% for every increase in the value of x by 1, $p = 45$. Substituting 14 for a and 45 for p in the equation $qx = a1 - \frac{p}{100}^x$ yields $qx = 141 - \frac{45}{100}^x$, which is equivalent to $qx = 141 - 0.45^x$, or $qx = 140.55^x$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. For this function, the value of qx increases, rather than decreases, by 45% for every increase in the value of x by 1.

▼ 2.19 — D

Choice D is correct. It's given that the equation $h = -4.9t^2 + 7t + 9$ represents this situation, where h is the height, in meters, of the object t seconds after it is kicked. It follows that the height, in meters, from which the object was kicked is the value of h when $t = 0$. Substituting 0 for t in the equation $h = -4.9t^2 + 7t + 9$ yields $h = -4.90^2 + 70 + 9$, or $h = 9$. Therefore, the object was kicked from a height of 9 meters.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

▼ 2.20 — B

Choice B is correct. It's given that the function $f(x) = 2061.034^x$ models the value, in dollars, of a certain bank account by the end of each year from 1957 through 1972, where x is the number of years after 1957. It follows that $f(x)$ represents the estimated value, in dollars, of the bank account x years after 1957. Since the value of $f(5)$ is the value of $f(x)$ when $x = 5$, it follows that "$f(5)$ is approximately equal to 243" means that $f(x)$ is approximately equal to 243 when $x = 5$. In the given context, this means that the value of the bank account is estimated to be approximately 243 dollars 5 years after 1957. Therefore, the best interpretation of the statement "$f(5)$ is approximately equal to 243" in this context is the value of the bank account is estimated to be approximately 243 dollars in 1962.

Choice A is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

Choice D is incorrect and may result from conceptual errors.

▼ 2.21 — B

Choice B is correct. The number that's 8 more than x can be represented by the expression $x + 8$. It's given that the product of x and $x + 8$ is 180, so it follows that $x(x + 8) = 180$, or $x^2 + 8x = 180$. Subtracting 180 from each side of this equation yields $x^2 + 8x - 180 = 0$. Factoring the left-hand side of this equation yields $(x - 10)(x + 18) = 0$. Applying the zero product property to this equation yields two solutions: $x = 10$ and $x = -18$. Since x is a positive number, the value of x is 10.

Choice A is incorrect. If $x = 5$, the product of x and the number that's 8 more than x would be $5(13)$, or 65, not 180.

Choice C is incorrect. This is the value of the number that's 8 more than x, not the value of x.

Choice D is incorrect. If $x = 36$, the product of x and the number that's 8 more than x would be $36(44)$, or 1,584, not 180.

▼ 2.22 — D

Choice D is correct. It's given that the expression $w(w + 9)$ represents the area, in square centimeters, of a rectangular cutting board, where w is the width, in centimeters, of the cutting board. The area of a rectangle can be calculated by multiplying its length by its width.

It follows that the length, in centimeters, of the cutting board is represented by the expression $w + 9$.

Choice A is incorrect. This expression represents the area, in square centimeters, of the cutting board, not its length, in centimeters.

Choice B is incorrect. This expression represents the width, in centimeters, of the cutting board, not its length.

Choice C is incorrect. This is the difference between the length, in centimeters, and the width, in centimeters, of the cutting board, not its length, in centimeters.

▼ 2.23 — C

Choice C is correct. The function f is quadratic, so it will have either a maximum or a minimum at the vertex of the graph. Since the coefficient of the quadratic term (–0.28) is negative, the vertex will be at a maximum. The equation $f(\theta) = -0.28(\theta - 27)^2 + 880$ is given in vertex form, so the vertex is at $\theta = 27$. Therefore, an angle of 27 degrees gives the greatest airflow.

Choices A and B are incorrect and may be the result of misidentifying which value in a quadratic equation in vertex form represents the vertex. Choice D is incorrect. This choice identifies the maximum value of $f(\theta)$ rather than the value of θ for which $f(\theta)$ is maximized.

▼ 2.24 — D

Choice D is correct. The equation of a quadratic function can be written in the form $fx = ax - h^2 + k$, where a, h, and k are constants. It's given in the table that when $x = -1$, the corresponding value of fx is 10. Substituting -1 for x and 10 for fx in the equation $fx = ax - h^2 + k$ gives $10 = a{-1} - h^2 + k$, which is equivalent to $10 = a1 + 2h + h^2 + k$, or $10 = a + 2ah + ah^2 + k$. It's given in the table that when $x = 0$, the corresponding value of fx is 14. Substituting 0 for x and 14 for fx in the equation $fx = ax - h^2 + k$ gives $14 = a0 - h^2 + k$, or $14 = ah^2 + k$. It's given in the table that when $x = 1$, the corresponding value of fx is 20. Substituting 1 for x and 20 for fx in the equation $fx = ax - h^2 + k$ gives $20 = a1 - h^2 + k$, which is equivalent to $20 = a1 - 2h + h^2 + k$, or $20 = a - 2ah + ah^2 + k$. Adding $20 = a - 2ah + ah^2 + k$ to the equation $10 = a + 2ah + ah^2 + k$ gives $30 = 2a + 2ah^2 + 2k$. Dividing both sides of this equation by 2 gives $15 = a + ah^2 + k$. Since $14 = ah^2 + k$, substituting 14 for $ah^2 + k$ into the

equation $15 = a + ah^2 + k$ gives $15 = a + 14$. Subtracting 14 from both sides of this equation gives $a = 1$. Substituting 1 for a in the equations $14 = ah^2 + k$ and $20 = ah^2 - 2ah + a + k$ gives $14 = h^2 + k$ and $20 = 1 - 2h + h^2 + k$, respectively. Since $14 = h^2 + k$, substituting 14 for $h^2 + k$ in the equation $20 = 1 - 2h + h^2 + k$ gives $20 = 1 - 2h + 14$, or $20 = 15 - 2h$. Subtracting 15 from both sides of this equation gives $5 = -2h$. Dividing both sides of this equation by -2 gives $-\frac{5}{2} = h$. Substituting $-\frac{5}{2}$ for h into the equation $14 = h^2 + k$ gives $14 = -\frac{5}{2}^2 + k$, or $14 = \frac{25}{4} + k$. Subtracting $\frac{25}{4}$ from both sides of this equation gives $\frac{31}{4} = k$. Substituting 1 for a, $-\frac{5}{2}$ for h, and $\frac{31}{4}$ for k in the equation $fx = ax - h^2 + k$ gives $fx = x + \frac{5}{2}^2 + \frac{31}{4}$, which is equivalent to $fx = x^2 + 5x + \frac{25}{4} + \frac{31}{4}$, or $fx = x^2 + 5x + 14$. Therefore, $fx = x^2 + 5x + 14$ defines f.

Choice A is incorrect. If $fx = 3x^2 + 3x + 14$, then when $x = -1$, the corresponding value of fx is 14, not 10.

Choice B is incorrect. If $fx = 5x^2 + x + 14$, then when $x = -1$, the corresponding value of fx is 18, not 10.

Choice C is incorrect. If $fx = 9x^2 - x + 14$, then when $x = -1$, the corresponding value of fx is 24, not 10, and when $x = 1$, the corresponding value of fx is 22, not 20.

▼ 2.25 — C

Choice C is correct. It's given that on April 1, there were 233 views of the advertisement. It's also given that every 2 days after April 1, the number of views of the advertisement had increased by 70% of the number of views 2 days earlier. This situation can be represented by an exponential function of the form $fx = a1 + \frac{r}{100}^{\frac{x}{c}}$, where a is the number of views on April 1 and every c days after April 1, the number of views had increased by $r\%$ of the number of views c days earlier. It follows that $a = 233$, $r = 70$, and $c = 2$. Substituting 233 for a, 70 for r, and 2 for c in the equation $fx = a1 + \frac{r}{100}^{\frac{x}{c}}$ yields $fx = 2331 + \frac{70}{100}^{\frac{x}{2}}$, or $fx = 2331.70^{\frac{x}{2}}$.

Choice A is incorrect. This function gives the predicted number of views for an advertisement for which every 2 days, the number of views was 70%, rather than increased by 70%, of the number of views 2 days earlier.

Choice B is incorrect. This function gives the predicted number of views for an advertisement for which every $\frac{1}{2}$ days, the number of views was 70% of the number of views $\frac{1}{2}$ days earlier, rather than an advertisement for which every 2 days, the number of views had increased by 70% of the number of views 2 days earlier.

Choice D is incorrect. This function gives the predicted number of views for an advertisement for which every $\frac{1}{2}$ days, rather than every 2 days, the number of views had increased by 70%

of the number of views $\frac{1}{2}$ days earlier, rather than 2 days earlier.

▼ 2.26 — C

Choice C is correct. It's given that the expression $ww + 29$ represents the area, in square centimeters, of a rectangular painting, where w is the width, in centimeters, of the painting. The area of a rectangle can be calculated by multiplying its length by its width. It follows that the length, in centimeters, of the painting is represented by the expression $w + 29$.

Choice A is incorrect. This expression represents the width, in centimeters, of the painting, not its length, in centimeters.

Choice B is incorrect. This is the difference between the length, in centimeters, and the width, in centimeters, of the painting, not its length, in centimeters.

Choice D is incorrect. This expression represents the area, in square centimeters, of the painting, not its length, in centimeters.

▼ 2.27 — B

Choice B is correct. It's given that the table shows three values of x and their corresponding values of y for the equation $y = 42^x + 3$. It's also given that when $x = 3$ the corresponding value of y is a, and a is a constant. Substituting 3 for x and a for y in the given equation yields $a = 42^3 + 3$, or $a = 35$. Therefore, the value of a is 35.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 2.28 — ?

The correct answer is 2.5. After hitting the ground once, the ball bounces to $20.0 \div 2 = 10.0$ feet. After hitting the ground a second time, the ball bounces to $10.0 \div 2 = 5.0$ feet. After hitting the ground for the third time, the ball bounces to $5.0 \div 2 = 2.5$ feet. Note that 2.5 and 5/2 are examples of ways to enter a correct answer.

▼ 2.29 — B

Choice B is correct. Let x be the first integer and let y be the second integer. If the first integer is 11 greater than twice the second integer, then $x = 2y + 11$. If the product of the two integers is 546, then $xy = 546$. Substituting $2y + 11$ for x in this equation results in $2y + 11y = 546$. Distributing the y to both terms in the parentheses results in $2y^2 + 11y = 546$. Subtracting 546 from both sides of this equation results in $2y^2 + 11y - 546 = 0$. The left-hand side of this equation can be factored by finding two values whose product is $2\text{-}546$, or $-1{,}092$, and whose sum is 11. The two values whose product is $-1{,}092$ and whose sum is 11 are 39 and -28. Thus, the equation $2y^2 + 11y - 546 = 0$ can be rewritten as $2y^2 + 28y - 39y - 546 = 0$, which is equivalent to $2yy - 14 + 39y - 14 = 0$, or $2y + 39y - 14 = 0$. By the zero product property, it follows that $2y + 39 = 0$ and $y - 14 = 0$. Subtracting 39 from both sides of the equation $2y + 39 = 0$ yields $2y = -39$. Dividing both sides of this equation by 2 yields $y = -\frac{39}{2}$. Since y is a positive integer, the value of y is not $-\frac{39}{2}$. Adding 14 to both sides of the equation $y - 14 = 0$ yields $y = 14$. Substituting 14 for y in the equation $xy = 546$ yields $x14 = 546$. Dividing both sides of this equation by 14 results in $x = 39$. Therefore, the two integers are 14 and 39, so the smaller of the two integers is 14.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the larger of the two integers.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 2.30 — B

Choice B is correct. It's given that the formula $P = C2^{rt}$ gives the number of bacteria in a growth medium, where r and C are constants and P is the number of bacteria t hours after the initial measurement. It's also given that a scientist initially measures $12{,}000$ bacteria in the growth medium. Since the initial measurement is 0 hours after the initial measurement, it follows that when $t = 0$, $P = 12{,}000$. Substituting 0 for t and $12{,}000$ for P in the given equation yields $12{,}000 = C2^{r0}$, or $12{,}000 = C2^0$, which is equivalent to $12{,}000 = C$. It's given that 4 hours later, the scientist measures $24{,}000$ bacteria, or when $t = 4$, $P = 24{,}000$. Substituting 4 for t, $24{,}000$ for P, and $12{,}000$ for C in the given equation yields $24{,}000 = 12{,}0002^{4r}$. Dividing each side of this equation by $12{,}000$ yields $2 = 2^{4r}$, or $2^1 = 2^{4r}$, which is equivalent to $1 = 4r$. Dividing both sides of this equation by 4 yields $\frac{1}{4} = r$. Therefore, the value of r is $\frac{1}{4}$.

Choice A is incorrect. This is the value of the reciprocal of C.

Choice C is incorrect. This is the value of the reciprocal of r.

Choice D is incorrect. This is the value of C.

▼ 2.31 — D

Choice D is correct. Substituting 0 for x in the given equation yields $f0 = 3{,}0000.75^0$, which is equivalent to $f0 = 3{,}0001$, or $f0 = 3{,}000$. It's given that the function estimates the species' population x years after 2008, so it follows that the estimated population of the species is $3{,}000$ in 2008. Therefore, the best interpretation of $3{,}000$ in this context is the estimated initial population for this species and area in 2008.

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect. The estimated percent decrease in the population for this species and area each year after 2008 is 25%, not $3{,}000$.

Choice C is incorrect. The estimated population for this species and area 8 years after 2008 is $3{,}0000.75^8$, or approximately 300, not $3{,}000$.

▼ 2.32 — B

Choice B is correct. For a quadratic function defined by an equation of the form $px = ax - h^2 + k$, where a, h, and k are constants and $a > 0$, the minimum value of the function is k. Subtracting 57 from both sides of the given equation yields $px = x^2 - 57$. This function is in the form $px = ax - h^2 + k$, where $a = 1$, $h = 0$, and $k = -57$. Therefore, the minimum value of the function p is -57.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 2.33 — B

Choice B is correct. A model for a quantity C that decreases by a certain percentage per time period t is an exponential equation in the form $C = I\left(1 - \frac{r}{100}\right)^t$, where I is the initial value at time $t = 0$ for r% annual decline. It's given that C is the number of millions of CDs sold in the United States and that t is the number of years after 2005. It's also given that 600 million

CDs were sold at time $t = 0$, so $I = 600$. This number declines by 15% per year, so $r = 15$. Substituting these values into the equation produces $C = 600\left(1 - \frac{15}{100}\right)^t$, or $C = 600(0.85)^t$.

Choice A is incorrect and may result from errors made when representing the percent decline. Choices C and D are incorrect. These equations model exponential increases in CD sales, not exponential decreases.

▼ 2.34 — D

Choice D is correct. It's given that the graph of the rational function *f* is shown, where $y = fx$ and $x \geq 0$. The graph shown passes through the point $3, 3$. It follows that when the value of x is 3, the value of fx is 3. When the value of fx is 3, the value of $fx + 5$ is $3 + 5$, or 8. Therefore, the graph of $y = fx + 5$ passes through the point $3, 8$. Of the given choices, choice D is the only graph that passes through the point $3, 8$ and is therefore the graph of $y = fx + 5$.

Choice A is incorrect. This is the graph of $y = fx - 5$, rather than $y = fx + 5$.

Choice B is incorrect. This is the graph of $y = \frac{fx}{5}$, rather than $y = fx + 5$.

Choice C is incorrect and may result from conceptual or calculation errors.

▼ 2.35 — B

Choice B is correct. It's given that t minutes after an initial observation, the number of bacteria in a population is $40{,}0002^{\frac{t}{790}}$. This expression consists of the initial number of bacteria, $40{,}000$, multiplied by the expression $2^{\frac{t}{790}}$. The time, in minutes, it takes for the number of bacteria to double is the increase in the value of t that causes the expression $2^{\frac{t}{790}}$ to double. Since the base is 2, the expression $2^{\frac{t}{790}}$ will double when the exponent increases by 1. Since the exponent of this expression is $\frac{t}{790}$, the exponent will increase by 1 when t increases by 790. Therefore, the time, in minutes, it takes for the number of bacteria in the population to double is 790.

Choice A is incorrect. This is the base of the exponent, not the time it takes for the number of bacteria in the population to double.

Choice C is incorrect. This is the number of minutes it takes for the population to double twice.

Choice D is incorrect. This is the number of bacteria that are initially observed, not the time it takes for the number of bacteria in the population to double.

▼ 2.36 — B

Choice B is correct. According to the table, $f(24) = -8$, $f(30) = -8$, and $f(32) = 8$. Let the quadratic function g be defined by $g(x) = f(x) + 8$. Therefore, $g(24) = f(24) + 8$, $g(30) = f(30) + 8$, and $g(32) = f(32) + 8$. Substituting -8 for $f(24)$ in the equation $g(24) = f(24) + 8$ yields $g(24) = -8 + 8$, or $g(24) = 0$. Substituting -8 for $f(30)$ in the equation $g(30) = f(30) + 8$ yields $g(30) = -8 + 8$, or $g(30) = 0$. Substituting 8 for $f(32)$ in the equation $g(32) = f(32) + 8$ yields $g(32) = 8 + 8$, or $g(32) = 16$. By the factor theorem, if $g(24) = 0$ and $g(30) = 0$, then $g(x) = k(x - 24)(x - 30)$, where k is a constant. Since $g(32) = 16$, substituting 32 for x and 16 for $g(x)$ in the equation $g(x) = k(x - 24)(x - 30)$ yields $16 = k(32 - 24)(32 - 30)$, or $16 = k(8)(2)$, which is equivalent to $16 = 16k$. Dividing both sides of this equation by 16 yields $1 = k$. Substituting 1 for k in the equation $g(x) = k(x - 24)(x - 30)$ yields $g(x) = (x - 24)(x - 30)$. Since $g(x) = f(x) + 8$, it follows that $g(x) - 8 = f(x)$. Subtracting 8 from both sides of the equation $g(x) = (x - 24)(x - 30)$ yields $g(x) - 8 = (x - 24)(x - 30) - 8$. Substituting $f(x)$ for $g(x) - 8$ in this equation yields $f(x) = (x - 24)(x - 30) - 8$.

Choice A is incorrect. If this equation defined f, then when $x = 24$, the value of $f(x)$ would be 4, not -8.

Choice C is incorrect. If this equation defined f, then when $x = 32$, the value of $f(x)$ would be 32, not 8.

Choice D is incorrect. If this equation defined f, then when $x = 32$, the value of $f(x)$ would be -32, not 8.

▼ 2.37 — A

Choice A is correct. For an exponential function of the form $Et = ab^t$, where a and b are constants, the initial value of the function—that is, the value of the function when $t = 0$—is a and the value of the function increases by a factor of b each time t increases by 1. Since the function $Et = 51.8^t$ gives the estimated number of employees at a restaurant and t is the number of years since the restaurant opened, the best interpretation of the number 5 in this context is the estimated number of employees when $t = 0$, or when the restaurant opened.

Choice B is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

Choice D is incorrect and may result from conceptual errors.

▼ 2.38 — A

Choice A is correct. The value of $f0$ is the value of fx when $x = 0$. The function f is defined by $fx = 10x^2 - 32x - 152$. Substituting 0 for x in this equation yields $f0 = 100^2 - 320 - 152$. This equation can be rewritten as $f0 = 100 - 0 - 152$, or $f0 = -152$. Therefore, the value of $f0$ is -152.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 2.39 — D

Choice D is correct. It's given that the function $fx = 651.03^x$ gives the predicted value, in dollars, of a certain rare coin x years after Immanuel purchased it. It follows that fx represents the predicted value, in dollars, of the coin x years after Immanuel purchased it. Since the value of $f8$ is the value of fx when $x = 8$, it follows that "$f8$ is approximately equal to 82" means that fx is approximately equal to 82 when $x = 8$. Therefore, the best interpretation of the statement "$f8$ is approximately equal to 82" in this context is 8 years after Immanuel purchased the rare coin, its predicted value is approximately 82 dollars.

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

▼ 2.40 — A

Choice A is correct. It's given that a rectangle has a length that is 15 times its width. It's also given that the function $y = 15ww$ represents this situation, where y is the area, in square feet, of the rectangle and $y > 0$. The area of a rectangle can be calculated by multiplying the rectangle's length by its width. Since the rectangle has a length that is 15 times its width, it follows that w represents the width of the rectangle, in feet, and $15w$ represents the length of the rectangle, in feet. Therefore, the best interpretation of $15w$ in this context is that it's the length of the rectangle, in feet.

Choice B is incorrect. This is the best interpretation of y, not $15w$, in the given function.

Choice C is incorrect and may result from conceptual errors.

Choice D is incorrect. This is the best interpretation of w, not $15w$, in the given function.

▼ 2.41 — A

Choice A is correct. A y-intercept is the point in the xy-plane where the graph of the function crosses the y-axis, which is where $x = 0$. It's given that the y-coordinate of the y-intercept of the graph of function f is c. It follows that the coordinate pair representing the y-intercept must be $(0,c)$. Therefore, c must equal $f(0)$.

Choices B, C, and D are incorrect because $f(1)$, $f(2)$, and $f(3)$would represent the y-value of the coordinate where $x = 1$, $x = 2$, and $x = 3$, respectively.

▼ 2.42 — A

Choice A is correct. It's given that $y = fx$. The x-intercepts of a graph in the xy-plane are the points where $y = 0$. Thus, for an x-intercept of the graph of function f, $0 = fx$. Substituting 0 for fx in the equation $fx = x^2 - 18x - 360$ yields $0 = x^2 - 18x - 360$. Factoring the right-hand side of this equation yields $0 = x + 12x - 30$. By the zero product property, $x + 12 = 0$ and $x - 30 = 0$. Subtracting 12 from both sides of the equation $x + 12 = 0$ yields $x = -12$. Adding 30 to both sides of the equation $x - 30 = 0$ yields $x = 30$. Therefore, the x-intercepts of the graph of $y = fx$ are $-12, 0$ and $30, 0$. Of these two x-intercepts, only $-12, 0$ is given as a choice.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 2.43 — 18

The correct answer is 18. It's given that an egg is thrown from a rooftop and that the equation $h = -4.9t^2 + 9t + 18$ represents this situation, where h is the height of the egg above the ground, in meters, t seconds after it is thrown. If follows that the height, in meters, from which the egg was thrown is the value of h when $t = 0$. Substituting 0 for t in

the equation $h = -4.9t^2 + 9t + 18$ yields $h0 = -4.90^2 + 90 + 18$, or $h = 18$. Therefore, according to the equation, the height, in meters, from which the egg was thrown is 18.

▼ 2.44 — D

Choice D is correct. It's given that a model predicts the population of Bergen in 2005 was $15{,}000$. The model also predicts that each year for the next 5 years, the population increased by 4% of the previous year's population. The predicted population in one of these years can be found by multiplying the predicted population from the previous year by 1.04. Since the predicted population in 2005 was $15{,}000$, the predicted population 1 year later is $15{,}000(1.04)$. The predicted population 2 years later is this value times 1.04, which is $15{,}0001.041.04$, or $15{,}0001.04^2$. The predicted population 3 years later is this value times 1.04, or $15{,}0001.04^3$. More generally, the predicted population, p, x years after 2005 is represented by the equation $p = 15{,}0001.04^x$.

Choice A is incorrect. Substituting 0 for x in this equation indicates the predicted population in 2005 was 0.96 rather than $15{,}000$.

Choice B is incorrect. Substituting 0 for x in this equation indicates the predicted population in 2005 was 1.04 rather than $15{,}000$.

Choice C is incorrect. This equation indicates the predicted population is decreasing, rather than increasing, by 4% each year.

▼ 2.45 — C

Choice C is correct. It's given that function f is defined by the equation $fx = 4x^{-1}$. The value of $f21$ is the value of fx when $x = 21$. Substituting 21 for x in the given equation yields $f21 = 421^{-1}$, which is equivalent to $f21 = 4\frac{1}{21}$, or $f21 = \frac{4}{21}$.

Choice A is incorrect. This is the value of $f21$ when $fx = -4x$, rather than $fx = 4x^{-1}$.

Choice B is incorrect. This is the value of $f21$ when $fx = 4x^{-1}$, rather than $fx = 4x^{-1}$.

Choice D is incorrect. This is the value of $f21$ when $fx = 4^{-1}x$, rather than $fx = 4x^{-1}$.

▼ 2.46 — B

Choice B is correct. The vertex of the graph of a quadratic equation is where it reaches its minimum or maximum value. When a quadratic equation is written in the form $y = a(x - h)^2 + k$, the vertex of the parabola represented by the equation is $(x, y) = (h, k)$. In

the given equation $y = -16(x - 5.6)^2 + 502$, the value of h is 5.6 and the value of k is 502. It follows that the vertex of the graph of this equation in the xy-plane is $(x, y) = (5.6, 502)$. Additionally, since $a = -16$ in the given equation, the graph of the quadratic equation opens down, and the vertex represents a maximum. It's given that the value of y represents the estimated height, in feet, of the toy rocket x seconds after it is launched into the air. Therefore, this toy rocket reaches an estimated maximum height of 502 feet 5.6 seconds after it is launched into the air.

Choice A is incorrect. The 16 in the equation is an indicator of how narrow the graph of the equation is rather than where it reaches its maximum.

Choice C is incorrect. The 16 in the equation is an indicator of how narrow the graph of the equation is rather than where it reaches its maximum.

Choice D is incorrect. This is an interpretation of the vertex of the graph of the equation $y = -16(x - 502)^2 + 5.6$, not of the equation $y = -16(x - 5.6)^2 + 502$.

▼ 2.47 — 11/4, 2.75

The correct answer is $\frac{11}{4}$. It's given that the function f is defined by $fx = 5\frac{1}{4} - x^2 + \frac{11}{4}$. Substituting $\frac{1}{4}$ for x in this equation yields $f\frac{1}{4} = 5\frac{1}{4} - \frac{1}{4}^2 + \frac{11}{4}$, which is equivalent $f\frac{1}{4} = 50^2 + \frac{11}{4}$, or $f\frac{1}{4} = \frac{11}{4}$. Therefore, the value of $f\frac{1}{4}$ is $\frac{11}{4}$. Note that 11/4 or 2.75 are examples of ways to enter a correct answer.

▼ 2.48 — C

Choice C is correct. If a value of x satisfies $f(x) = 0$, the graph of $y = fx$ will contain a point $x, 0$ and thus touch the x-axis. Since there are 3 points at which this graph touches the x-axis, there are 3 values of x for which $f(x) = 0$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 2.49 — A

Choice A is correct. The y-intercept of a graph is the point at which the graph intersects the y-axis. The graph of $y = fx$ shown intersects the y-axis at $0, 3$, so the y-intercept of the graph of $y = fx$ is $0, 3$. The graph of $y = fx + 5$ is the result of shifting the graph of $y = fx$ up 5 units.

Therefore, the *y*-intercept of the graph of $y = fx + 5$ is 5 units above the *y*-intercept of the graph of $y = f(x)$. It follows that the *y*-intercept of the graph of $y = fx + 5$ is $0, 3 + 5$, which is equivalent to $0, 8$.

Choice B is incorrect. The point $0, 5$ is 5 units above the point $0, 0$, not the point $0, 3$.

Choice C is incorrect. This is the *y*-intercept of the graph of $y = fx$, not the *y*-intercept of the graph of $y = fx + 5$.

Choice D is incorrect. This is the origin of the coordinate plane, not the *y*-intercept of the graph of $y = fx + 5$.

▼ 2.50 — A

Choice A is correct. The value 1.4 is the value of x, which represents the number of seconds after the object was thrown straight up in the air. When the function h is evaluated for x = 1.4, the function has a value of 1.64, which is the height, in feet, of the object.

Choices B and C are incorrect and may result from misidentifying seconds as feet and feet as seconds. Additionally, choice C may result from incorrectly including the initial height of the object as the input x. Choice D is incorrect and may result from misidentifying height as speed.

▼ 2.51 — D

Choice D is correct. The y-intercept of the graph of an equation is the point $(0,b)$, where b is the value of y when $x = 0$. For the given equation, $y = 4$ when $x = 0$. It follows that the y-intercept of the graph of the given equation is $(0,4)$. Additionally, for the given equation, the value of y doubles for each increase of 1 in the value of x. Therefore, the graph contains the points $(1,8)$, $(2,16)$, $(3,32)$, and $(4,64)$. Only the graph shown in choice D passes through these points.

Choices A and B are incorrect because these are graphs of decreasing, not increasing, exponential functions. Choice C is incorrect because the value of y increases by a growth factor greater than 2 for each increase of 1 in the value of x.

▼ 2.52 — B

Choice B is correct. The area of a rectangle is equal to its length multiplied by its width. Multiplying the given length, x units, by the given width, $x - 15$ units, yields $xx - 15$ square units. If the rectangle has an area of 76 square units, it follows that $xx - 15 = 76$, or $x^2 - 15x = 76$. Subtracting 76 from both sides of this equation yields $x^2 - 15x - 76 = 0$. Factoring the left-hand side of this equation yields $x - 19x + 4 = 0$. Applying the zero product property to this equation yields two solutions: $x = 19$ and $x = -4$. Since x is the rectangle's length, in units, which must be positive, the value of x is 19.

Choice A is incorrect. This is the width, in units, of the rectangle, not the value of x.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the area, in square units, of the rectangle, not the value of x.

▼ 2.53 — D

Choice D is correct. The y-intercept of the graph is the point at which the graph crosses the y-axis, or the point for which the value of x is 0. Therefore, the y-intercept of the given graph is the point $0, 9$. It's given that x represents the number of years since the end of 1992. Therefore, $x = 0$ represents 0 years since the end of 1992, which is the same as the end of 1992. It's also given that y represents the estimated number of catalogs, in thousands, that the company sent to its customers at the end of the year. Therefore, $y = 9$ represents $9{,}000$ catalogs. It follows that the y-intercept $0, 9$ means that the estimated number of catalogs the company sent to its customers at the end of 1992 was $9{,}000$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

▼ 2.54 — D

Choice D is correct. Each choice has a function with coefficient 5 and base 2, so the exponents must be analyzed. When the input value of x increases, the output value of f(x) decreases, so the exponent must be negative. An exponent of –x yields the values in the table: $5 = 5(2^{-0})$, $\frac{5}{2} = 5(2^{-1})$, $\frac{5}{4} = (2^{-2})$, and $\frac{5}{8} = 5(2^{-3})$.

Choices A and B are incorrect and may result from choosing equations that yield an increasing, rather than decreasing, output value of f(x) when the input value of x increases.

Choice C is incorrect and may result from choosing an equation that doesn't yield the values in the table.

▼ 2.55 — 68

The correct answer is 68. It's given that the function f is defined by $fx = 8x^3 + 4$. Substituting 2 for x in this equation yields $f2 = 82^3 + 4$, or $f2 = 88 + 4$, which is equivalent to $f2 = 68$. Therefore, the value of $f2$ is 68.

▼ 2.56 — A

Choice A is correct. It's given that the function f models the intensity of an X-ray beam, in number of particles in the X-ray beam, t millimeters below the surface of a sample of iron. When the X-ray beam is at the surface of the sample of iron, it is 0 millimeters below the surface, so the value of t is 0. Substituting 0 for t in the function $ft = 5000.5^{\frac{t}{12}}$ yields $f0 = 5000.5^{\frac{0}{12}}$. Since any positive number raised to the power of 0 is equal to 1, it follows that $f0 = 5001$, or $f0 = 500$. Therefore, the estimated number of particles in the X-ray beam at the surface of the sample of iron is 500.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 2.57 — 17

The correct answer is 17. It's given that the graph of $y = f(x)$ in the xy-plane passes through the point $(0, b)$, where b is a constant. It follows that $f(0)$ equals b. Substituting 0 for x in the given function yields $f(0) = 0^5 + 9(0) + 17$, or $f(0) = 17$. Therefore, the value of b is 17.

▼ 2.58 — B

Choice B is correct. It's given that the relationship between t and d is exponential. The table shows that the value of d increases as the value of t increases. Therefore, the relationship between t and d can be represented by an increasing exponential equation of the form $d = a1 + b^t$, where a and b are positive constants. The table shows that when $t = 0$, $d = 670$. Substituting 0 for t and 670 for d in the equation $d = a1 + b^t$ yields $670 = a1 + b^0$, which is equivalent to $670 = a1$, or $670 = a$. Substituting 670 for a in the equation $d = a1 + b^t$ yields

$d = 6701 + b^t$. The table also shows that when $t = 1$, $d = 674.02$. Substituting 1 for t and 674.02 for d in the equation $d = 6701 + b^t$ yields $674.02 = 6701 + b^1$, or $674.02 = 6701 + b$. Dividing both sides of this equation by 670 yields $1.006 = 1 + b$. Subtracting 1 from both sides of this equation yields $b = 0.006$. Substituting 0.006 for b in the equation $d = 6701 + b^t$ yields $d = 6701 + 0.006^t$. Therefore, of the choices, choice B best represents the relationship between t and d.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 2.59 — B

Choice B is correct. It's given that the function h estimates that the object is $3{,}364$ feet above the ground when it's dropped at $t = 0$. Substituting $3{,}364$ for ht and 0 for t in the function h yields $3{,}364 = -160^2 + b$, or $3{,}364 = b$. Substituting $3{,}364$ for b in the function h yields $ht = -16t^2 + 3{,}364$. When the object hits the ground, its height will be 0 feet above the ground. Substituting 0 for ht in $ht = -16t^2 + 3{,}364$ yields $0 = -16t^2 + 3{,}364$. Adding $16t^2$ to each side of this equation yields $16t^2 = 3{,}364$. Dividing each side of this equation by 16 yields $t^2 = 210.25$. Since the object will hit the ground at a positive number of seconds after it's dropped, the value of t can be found by taking the positive square root of each side of this equation, which yields $t = 14.50$. It follows that the function estimates the object will hit the ground approximately 14.50 seconds after being dropped.

Choice A is incorrect. The function estimates that 7.25 seconds after being dropped, the object's height will be $-167.25^2 + 3{,}364$ feet, or $2{,}523$ feet, above the ground.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 2.60 — C

Choice C is correct. A model for a quantity S that increases by a certain percentage per time period n is an exponential function in the form $S(n) = I\left(1 + \frac{r}{100}\right)^n$, where I is the initial value at time $n = 0$ for r% annual increase. It's given that the annual increase in an employee's salary is 4%, so $r = 4$. The initial value can be found by substituting 0 for n in the

given function, which yields $S(0) = 38{,}000$. Therefore, $I = 38{,}000$. Substituting these values for r and I into the form of the exponential function $S(n) = I\left(1 + \frac{r}{100}\right)^n$ yields $S(n) = 38{,}000\left(1 + \frac{4}{100}\right)^n$, or $S(n) = 38{,}000(1.04)^n$. Therefore, the value of a in the given function is 1.04.

Choices A, B, and D are incorrect and may result from incorrectly representing the annual increase in the exponential function.

▼ 2.61 — ?

Choice B is correct. Let x be the number of years after 1985. It follows that $\frac{x}{2}$ represents the number of 2-year periods that will occur within an x-year period. According to Moore's law, every 2 years, the number of transistors included on microprocessors is estimated to double. Therefore, x years after 1985, the number of transistors will double $\frac{x}{2}$ times. Since the number of transistors included on a microprocessor was 275,000, or .275 million, in 1985, the estimated number of transistors, in millions, included x years after 1985 can be modeled as $0.275 \cdot 2^{\frac{x}{2}}$. The year in which the number of transistors is estimated to be 1.1 million is represented by the value of x when $1.1 = 0.275 \cdot 2^{\frac{x}{2}}$. Dividing both sides of this equation by .275 yields $4 = 2^{\frac{x}{2}}$, which can be rewritten as $2^2 = 2^{\frac{x}{2}}$. Since the exponential equation has equal bases on each side, it follows that the exponents must also be equal: $2 = \frac{x}{2}$. Multiplying both sides of the equation $2 = \frac{x}{2}$ by 2 yields $x = 4$. Therefore, according to Moore's law, 4 years after 1985, or in 1989, the number of transistors included on microprocessors is estimated to reach 1.1 million.

Alternate approach: According to Moore's law, 2 years after 1985 (in 1987), the number of transistors included on a microprocessor is estimated to be $2 \cdot 275{,}000$, or 550,000, and 2 years after 1987 (in 1989), the number of transistors included on microprocessors is estimated to be $2 \cdot 550{,}000$, or 1,100,000. Therefore, the year that Moore's law estimates the number of transistors on microprocessors to reach 1.1 million is 1989.

Choices A, C, and D are incorrect. According to Moore's law, the number of transistors included on microprocessors is estimated to reach 550,000 in 1987, 2.2 million in 1991, and about 6.2 million in 1994.

▼ 2.62 — 11875

The correct answer is $11{,}875$. It's given that the exponential function g is defined by $gx = 19 \cdot a^x$, where a is a positive constant, and $g3 = 2{,}375$. It follows that when $x = 3$, $gx = 2{,}375$. Substituting 3 for x and $2{,}375$ for gx in the given equation yields $2{,}375 = 19 \cdot a^3$. Dividing each side of this equation by 19 yields $125 = a^3$. Taking the cube root of both sides of this equation gives $a = 5$. Substituting 4 for x and 5 for a in the equation $gx = 19 \cdot a^x$ yields $g4 = 19 \cdot 5^4$, or $g4 = 11{,}875$. Therefore, the value of $g4$ is $11{,}875$.

▼ 2.63 — B

Choice B is correct. On the graph shown, the y-axis represents estimated population, in thousands. The graph shows that when $x = 0$, the y-coordinate is 6. Therefore, the estimated population at $x = 0$ is 6 thousand. The graph also shows that when $x = 1$, the y-coordinate is 9. Therefore, the estimated population at $x = 1$ is 9 thousand. Dividing 9 thousand by 6 thousand yields 1.5; therefore, 9 thousand is 1.5 times 6 thousand. It follows that the estimated population at $x = 1$ is 1.5 times the estimated population at $x = 0$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 2.64 — B

Choice B is correct. The graph of the function f in the xy-plane has x-intercepts at the points (x,y), where $y = f(x) = 0$. Substituting 0 for $f(x)$ in the given equation yields $0 = (x+4)(x-1)(2x-3)$. By the zero product property, if $0 = (x+4)(x-1)(2x-3)$, then $x+4 = 0$, $x-1 = 0$, or $2x-3 = 0$. Solving each of these linear equations for x, it follows that $x = -4$, $x = 1$, and $x = \frac{3}{2}$, respectively. This means that the graph of the function f in the xy-plane has three x-intercepts: $(-4,0)$, $(1,0)$, and $\left(\frac{3}{2},0\right)$. Therefore, $\left(-\frac{2}{3},0\right)$ isn't an x-intercept of the graph of the function f.

Alternate approach: Substitution may be used. Since by definition an x-intercept of any graph

is a point in the form $(k,0)$ where k is a constant, and since all points in the options are in this form, it need only be checked whether the points in the options lie on the graph of the function f. Substituting $-\frac{2}{3}$ for x and 0 for $f(x)$ in the given equation yields $0=\left(-\frac{2}{3}+4\right)\left(-\frac{2}{3}-1\right)\left(2\left(-\frac{2}{3}\right)-3\right)$, or $0=\frac{650}{27}$. Therefore, the point $\left(-\frac{2}{3},0\right)$ doesn't lie on the graph of the function f and can't be an x-intercept of the graph.

Choices A, C, and D are incorrect because each of these points is an x-intercept of the graph of the function f in the xy-plane. By definition, an x-intercept is a point on the graph of the form $(k,0)$, where k is a constant. Substituting -4 for x and 0 for $f(x)$ in the given equation yields $0=(-4+4)(-4-1)(2(-4)-3)$, or $0=0$. Since this is a true statement, the point $(-4,0)$ lies on the graph of the function f and is an x-intercept of the graph. Performing similar substitution using the points $(1,0)$ and $\left(\frac{3}{2},0\right)$ also yields the true statement $0=0$, illustrating that these points also lie on the graph of the function f and are x-intercepts of the graph.

▼ 2.65 — A

Choice A is correct. It's given that an object's kinetic energy, in joules, is equal to the product of one-half the object's mass, in kilograms, and the square of the object's speed, in meters per second. This relationship can be represented by the equation $K=\frac{1}{2}mv^2$, where K is the kinetic energy, m is the mass, and v is the speed. Substituting a mass of 4 kilograms for m and a kinetic energy of 18 joules for K results in the equation $18=\left(\frac{1}{2}\right)(4)v^2$, or $18=2v^2$. Dividing both sides of this equation by 2 yields $9=v^2$. Taking the square root of both sides yields $v=-3$ and $v=3$. Since speed can't be expressed as a negative number, the speed of the object is 3 meters per second.

Choice B is incorrect and may result from computation errors. Choice C is incorrect. This is the value of v^2 rather than v. Choice D is incorrect. This is the value of $4v^2$ rather than v.

▼ 2.66 — A

Choice A is correct. It's given that the volleyball court is rectangular and has an area of 162 square meters. The formula for the area of a rectangle is $A = \ell \cdot w$, where A is the area, ℓ is the length, and w is the width of the rectangle. It's also given that the length of the volleyball court is twice the width, thus $\ell = 2w$. Substituting the given value into the formula for the area of a rectangle and using the relationship between length and width for this rectangle yields $162 = (2w)(w)$. This equation can be rewritten as $162 = 2w^2$. Dividing both sides of this equation by 2 yields $81 = w^2$. Taking the square root of both sides of this equation yields $\pm 9 = w$. Since the width of a rectangle is a positive number, the width of the volleyball court is 9 meters.

Choice B is incorrect because this is the length of the rectangle. Choice C is incorrect because this is the result of using 162 as the perimeter rather than the area. Choice D is incorrect because this is the result of calculating w in the equation $162 = 2w + w$ instead of $162 = (2w)(w)$.

▼ 2.67 — B

Choice B is correct. The area, A, of a triangle is given by the formula $A = \frac{1}{2}bh$, where b represents the length of the base of the triangle and h represents its height. It's given that the area of a triangle is 270 square centimeters and that the length of the base of this triangle is 12 centimeters greater than the height of the triangle. Let x represent the height, in centimeters, of the triangle. It follows that the length of the base of the triangle can be expressed as $x + 12$. Substituting 270 for A, x for h, and $x + 12$ for b in the formula $A = \frac{1}{2}bh$ yields $270 = \frac{1}{2}(x + 12)(x)$, or $270 = \frac{1}{2}x(x + 12)$. Multiplying both sides of this equation by 2 yields $540 = x(x + 12)$. Applying the distributive property on the right-hand side of this equation yields $540 = x^2 + 12x$. Subtracting 540 from both sides of this equation yields $0 = x^2 + 12x - 540$. In factored form, this equation is equivalent to $(x + 30)(x - 18) = 0$. Applying the zero product property, it follows that $x + 30 = 0$ or $x - 18 = 0$. Subtracting 30 from both sides of the equation $x + 30 = 0$ yields $x = -30$. Adding 18 to both sides of the equation $x - 18 = 0$ yields $x = 18$. Since x represents the height of the triangle, it must be positive. Therefore, the height, in centimeters, of the triangle is 18.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 2.68 — D

Choice D is correct. Substituting -1 for x in the given function yields $q(-1) = 32^{-1}$, which is equivalent to $q(-1) = 32^{\frac{1}{2}}$, or $q(-1) = 16$. Therefore, when $x = -1$, the corresponding value of qx for function q is 16. Substituting 0 for x in the given function yields $q0 = 32^0$, which is equivalent to $q0 = 321$, or $q0 = 32$. Therefore, when $x = 0$, the corresponding value of qx for function q is 32. Substituting 1 for x in the given function yields $q1 = 32^1$, which is equivalent to $q1 = 322$, or $q1 = 64$. Therefore, when $x = 1$, the corresponding value of qx for function q is 64. Of the choices given, only the table in choice D gives these three values of x and their corresponding values of qx for function q.

Choice A is incorrect. This table gives three values of x and their corresponding values of qx for the function $qx = 322x$.

Choice B is incorrect. This table gives three values of x and their corresponding values of qx for the function $qx = 232^x$.

Choice C is incorrect and may result from conceptual or calculation errors.

▼ 2.69 — D

Choice D is correct. The table shows an increasing exponential relationship between the number of years, x, since Hana started training in pole vault and the estimated height hx, in meters, of her best pole vault for that year. The relationship can be written as $hx = Ca^x$, where C and a are positive constants. It's given that when $x = 0$, $hx = 1.23$. Substituting 0 for x and 1.23 for hx in $hx = Ca^x$ yields $1.23 = Ca^0$, or $1.23 = C$. Substituting 1.23 for C in $hx = Ca^x$ yields $hx = 1.23a^x$. It's also given that when $x = 2$, $hx = 1.54$. Substituting 2 for x and 1.54 for hx in $hx = 1.23a^x$ yields $1.54 = 1.23a^2$. Dividing each side of this equation by 1.23 yields $\frac{1.54}{1.23} = \frac{1.23a^2}{1.23}$, or a^2 is approximately equal to 1.252. Since a is positive, a is approximately equal to $\sqrt{1.252}$, or 1.12. Substituting 1.12 for a in $hx = 1.23a^x$ yields $hx = 1.231.12^x$.

Choice A is incorrect. When $x = 0$, the value of hx in this function is equal to 1.12 rather than 1.23, and it is decreasing rather than increasing.

Choice B is incorrect. When $x = 0$, the value of hx in this function is equal to 1.12 rather than 1.23.

Choice C is incorrect. This function is decreasing rather than increasing.

▼ 2.70 — D

Choice D is correct. It's given that function g is exponential. Therefore, an equation defining g can be written in the form $gx = ab^x$, where a and b are constants. The table shows that when $x = 0$, $gx = 1$. Substituting 0 for x and 1 for gx in the equation $gx = ab^x$ yields $1 = ab^0$, which is equivalent to $1 = a$. Substituting 1 for a in the equation $gx = ab^x$ yields $gx = b^x$. The table also shows that when $x = 1$, $gx = \frac{1}{25}$. Substituting 1 for x and $\frac{1}{25}$ for gx in the equation $gx = b^x$ yields $\frac{1}{25} = b^1$, which is equivalent to $\frac{1}{25} = b$. Substituting $\frac{1}{25}$ for b in the equation $gx = b^x$ yields $gx = \frac{1}{25}^x$.

Choice A is incorrect. For this function, $g1$ is equal to -25, not $\frac{1}{25}$.

Choice B is incorrect. For this function, $g1$ is equal to $-\frac{1}{25}$, not $\frac{1}{25}$.

Choice C is incorrect. For this function, $g1$ is equal to 25, not $\frac{1}{25}$.

▼ 2.71 — B

Choice B is correct. It's given that $st = rt + 1$. Since $rt = 63t - 2t^2$, it follows that $st = 63t - 2t^2 + 1$, or $st = -2t^2 + 63t + 1$. For a quadratic function s defined by an equation of the form $st = at - h^2 + k$, where a, h, and k are constants and a is negative, the value of $s(t)$ reaches its maximum, k, when the value of t is h. The equation $st = -2t^2 + 63t + 1$ can be rewritten in the form $st = a(t - h)^2 + k$ by completing the square. The equation $s(t) = -2t^2 + 63t + 1$ is equivalent to $st = -2t^2 - \frac{63}{2}t + 1$, or $st = -2t^2 - \frac{63}{2}t + \frac{63}{4}^2 - \frac{63}{4}^2 + 1$. This equation can be rewritten as $st = -2t - \frac{63}{4}^2 - \frac{63}{4}^2 + 1$, or $st = -2t - \frac{63}{4}^2 + 1 + 2\frac{63}{4}^2$. This equation is in the form $st = at - h^2 + k$, where $a = -2$, $h = \frac{63}{4}$, and $k = 1 + 2\frac{63}{4}^2$. Therefore, the maximum value of st is $1 + 2\frac{63}{4}^2$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 2.72 — B

Choice B is correct. It's given that $fx = 94^x$ and $gx = fx + 2$. Substituting $x + 2$ for x in $fx = 94^x$ gives $fx + 2 = 94^{x+2}$. Rewriting this equation using properties of exponents gives

$f(x+2) = 9^{4x}4^2$, which is equivalent to $f(x+2) = 9^{4x}16$. Multiplying 9 and 16 in this equation gives $f(x+2) = 1444^x$. Since $g(x) = f(x+2)$, $g(x) = 1444^x$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 2.73 — D

Choice D is correct. The value of p(12) depends on the value of the constant c, so the value of c must first be determined. It is given that p(c) = 10. Based on the definition of p, it follows that:

$$p(c) = \frac{(c-c)^2+160}{2c} = 10$$

$$\frac{160}{2c} = 10$$

$$2c = 16$$

$$c = 8$$

This means that $p(x) = \frac{(x-8)^2+160}{16}$ for all values of x. Therefore:

$$p(12) = \frac{(12-8)^2+160}{16}$$

$$= \frac{16+160}{16}$$

$$= 11$$

Choice A is incorrect. It is the value of p(8), not p(12). Choices B and C are incorrect. If one of these values were correct, then x = 12 and the selected value of p(12) could be substituted into the equation to solve for c. However, the values of c that result from choices B and C each result in p(c) < 10.

▼ 2.74 — D

Choice D is correct. The y-intercept of a graph in the ty-plane is the point where $t = 0$. For the given function f, the y-intercept of the graph of $y = ft$ in the ty-plane can be found by substituting 0 for t in the equation $y = 8{,}0000 \cdot 65^t$, which gives $y = 8{,}0000 \cdot 65^0$. This is equivalent to $y = 8{,}0001$, or $y = 8{,}000$. Therefore, the y-intercept of the graph of $y = ft$ is $0, 8{,}000$. It's given that the function f models the number of coupons a company sent to their customers at the end of each year. Therefore, ft represents the estimated number of coupons the company sent to their customers at the end of each year. It's also given that t represents the number of years since the end of 1998. Therefore, $t = 0$ represents 0 years since the end of 1998, or the end of 1998. Thus, the best interpretation of the y-intercept of the graph of $y = ft$ is that the estimated number of coupons the company sent to their customers at the end of 1998 was $8{,}000$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

▼ 2.75 — D

Choice D is correct. It's given that the function p models the population of Lowell t years after a census. Since there are 12 months in a year, m months after the census is equivalent to $\frac{m}{12}$ years after the census. Substituting $\frac{m}{12}$ for t in the equation $pt = 90{,}0001.06^t$ yields $p\frac{m}{12} = 90{,}0001.06^{\frac{m}{12}}$. Therefore, the function r that best models the population of Lowell m months after the census is $rm = 90{,}0001.06^{\frac{m}{12}}$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

▼ 2.76 — 168

The correct answer is 168. The quadratic function g gives the estimated depth of the seal, gt, in meters, t minutes after the seal enters the water. It's given that function g estimates that the seal reached its maximum depth of 302.4 meters 6 minutes after it entered the water. Therefore, function g can be expressed in vertex form as $gt = at - 6^2 + 302.4$, where a is a constant. Since it's also given that the seal reached the surface of the water after 12 minutes, $g12 = 0$. Substituting 12 for t and 0 for gt in $gt = at - 6^2 + 302.4$ yields

$0 = a12 - 6^2 + 302.4$, or $36a = -302.4$. Dividing both sides of this equation by 36 gives $a = -8.4$. Substituting -8.4 for a in $gt = at - 6^2 + 302.4$ gives $gt = -8.4t - 6^2 + 302.4$. Substituting 10 for t in gt gives $g10 = -8.410 - 6^2 + 302.4$, which is equivalent to $g10 = -8.44^2 + 302.4$, or $g10 = 168$. Therefore, the estimated depth, to the nearest meter, of the seal 10 minutes after it entered the water was 168 meters.

▼ 2.77 — D

Choice D is correct. Since w represents the nth term of the sequence and 9 is the first term of the sequence, the value of w is 9 when the value of n is 1. Since each term after the first is 4 times the preceding term, the value of w is 94 when the value of n is 2. Therefore, the value of w is 944, or 94^2, when the value of n is 3. More generally, the value of w is 94^{n-1} for a given value of n. Therefore, the equation $w = 94^{n-1}$ gives w in terms of n.

Choice A is incorrect. This equation describes a sequence for which the first term is 36, rather than 9, and each term after the first is 9, rather than 4, times the preceding term.

Choice B is incorrect. This equation describes a sequence for which the first term is 4, rather than 9, and each term after the first is 9, rather than 4, times the preceding term.

Choice C is incorrect. This equation describes a sequence for which the first term is 36, rather than 9.

▼ 2.78 — C

Choice C is correct. It's given that the function P models the population of the city t years after 2005. Since there are 12 months in a year, 18 months is equivalent to $\frac{18}{12}$ years. Therefore, the expression $\frac{18}{12}x$ can represent the number of years in x 18-month periods. Substituting $\frac{18}{12}x$ for t in the given equation yields $P\frac{18}{12}x = 2901.04^{\frac{4}{6}\frac{18}{12}x}$, which is equivalent to $P\frac{18}{12}x = 2901.04^{x}$. Therefore, for each 18-month period, the predicted population of the city is 1.04 times, or 104% of, the previous population. This means that the population is predicted to increase by 4% every 18 months.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. Each year, the predicted population of the city is 1.04 times the previous year's predicted population, which is not the same as an increase of 1.04%.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 2.79 — A

Choice A is correct. The value of k for which $fk = 3k$ can be found by substituting k for x and $3k$ for fx in the given equation, $fx = 59 - 2x$, which yields $3k = 59 - 2k$. For this equation to be true, either $-3k = 59 - 2k$ or $3k = 59 - 2k$. Adding $2k$ to both sides of the equation $-3k = 59 - 2k$ yields $-k = 59$. Dividing both sides of this equation by -1 yields $k = -59$. To check whether -59 is the value of k, substituting -59 for k in the equation $3k = 59 - 2k$ yields $3\text{-}59 = 59 - 2\text{-}59$, which is equivalent to $-177 = 177$, or $-177 = 177$, which isn't a true statement. Therefore, -59 isn't the value of k. Adding $2k$ to both sides of the equation $3k = 59 - 2k$ yields $5k = 59$. Dividing both sides of this equation by 5 yields $k = \frac{59}{5}$. To check whether $\frac{59}{5}$ is the value of k, substituting $\frac{59}{5}$ for k in the equation $3k = 59 - 2k$ yields $3\frac{59}{5} = 59 - 2\frac{59}{5}$, which is equivalent to $\frac{177}{5} = \frac{177}{5}$, or $\frac{177}{5} = \frac{177}{5}$, which is a true statement. Therefore, the value of k for which $fk = 3k$ is $\frac{59}{5}$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 2.80 — -324

The correct answer is -324. It's given that the function f is defined by $fx = a^x + b$, where a and b are constants. It's also given that the graph of $y = fx$ has a y-intercept at $0, -323$. It follows that $f0 = -323$. Substituting 0 for x and -323 for fx in $fx = a^x + b$ yields $-323 = a^0 + b$, or $-323 = 1 + b$. Subtracting 1 from each side of this equation yields $-324 = b$. Therefore, the value of b is -324.

▼ 2.81 — -7

The correct answer is -7. For a quadratic function defined by an equation of the form $fx = ax - h^2 + k$, where a, h, and k are constants and $a > 0$, the function reaches its minimum when $x = h$. In the given function, $a = 1$, $h = -7$, and $k = 4$. Therefore, the value of x for which fx reaches its minimum is -7.

▼ 2.82 — D

Choice D is correct. It's given that $fx = x - 10x + 13$, which can be rewritten as $fx = x^2 + 3x - 130$. Since the coefficient of the x^2-term is positive, the graph of $y = fx$ in the xy-plane opens upward and reaches its minimum value at its vertex. The x-coordinate of the vertex is the value of x such that fx reaches its minimum. For an equation in the form $fx = ax^2 + bx + c$, where a, b, and c are constants, the x-coordinate of the vertex is $-\frac{b}{2a}$. For

the equation $fx = x^2 + 3x - 130$, $a = 1$, $b = 3$, and $c = -130$. It follows that the x-coordinate of the vertex is $-\frac{3}{21}$, or $-\frac{3}{2}$. Therefore, fx reaches its minimum when the value of x is $-\frac{3}{2}$.

Alternate approach: The value of x for the vertex of a parabola is the x-value of the midpoint between the two x-intercepts of the parabola. Since it's given that $fx = x - 10x + 13$, it follows that the two x-intercepts of the graph of $y = fx$ in the xy-plane occur when $x = 10$ and $x = -13$, or at the points $10, 0$ and $-13, 0$. The midpoint between two points, x_1, y_1 and x_2, y_2, is $\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}$. Therefore, the midpoint between $10, 0$ and $-13, 0$ is $\frac{10 + -13}{2}, \frac{0 + 0}{2}$, or $-\frac{3}{2}, 0$. It follows that fx reaches its minimum when the value of x is $-\frac{3}{2}$.

Choice A is incorrect. This is the y-coordinate of the y-intercept of the graph of $y = fx$ in the xy-plane.

Choice B is incorrect. This is one of the x-coordinates of the x-intercepts of the graph of $y = fx$ in the xy-plane.

Choice C is incorrect and may result from conceptual or calculation errors.

▼ 2.83 — C

Choice C is correct. It's given that $fx = \frac{a}{x + b}$ and that the graph shown is a partial graph of $y = fx$. Substituting y for fx in the equation $fx = \frac{a}{x + b}$ yields $y = \frac{a}{x + b}$. The graph passes through the point $-7, -2$. Substituting -7 for x and -2 for y in the equation $y = \frac{a}{x + b}$ yields $-2 = \frac{a}{-7 + b}$. Multiplying each side of this equation by $-7 + b$ yields $-2-7 + b = a$, or $14 - 2b = a$. The graph also passes through the point $-5, -6$. Substituting -5 for x and -6 for y in the equation $y = \frac{a}{x + b}$ yields $-6 = \frac{a}{-5 + b}$. Multiplying each side of this equation by $-5 + b$ yields $-6-5 + b = a$, or $30 - 6b = a$. Substituting $14 - 2b$ for a in this equation yields $30 - 6b = 14 - 2b$. Adding $6b$ to each side of this equation yields $30 = 14 + 4b$. Subtracting 14 from each side of this equation yields $16 = 4b$. Dividing each side of this equation by 4 yields $4 = b$. Substituting 4 for b in the equation $14 - 2b = a$ yields $14 - 24 = a$, or $6 = a$. Substituting 6 for a and 4 for b in the equation $fx = \frac{a}{x + b}$ yields $fx = \frac{6}{x + 4}$. It's given that $gx = fx + 4$. Substituting $x + 4$ for x in the equation $fx = \frac{6}{x + 4}$ yields $fx + 4 = \frac{6}{x + 4 + 4}$, which is equivalent to $fx + 4 = \frac{6}{x + 8}$. It follows that $gx = \frac{6}{x + 8}$.

Choice A is incorrect. This could define function g if $gx = fx - 4$.

Choice B is incorrect. This could define function g if $gx = fx$.

Choice D is incorrect. This could define function g if $gx = fx \cdot x + 4$.

▼ 2.84 — D

Choice D is correct. It's given that the graph of $y = fx$ in the xy-plane is a parabola with vertex h, k. If $f\text{-}9 = f3$, then for the graph of $y = fx$, the point with an x-coordinate of -9 and the point with an x-coordinate of 3 have the same y-coordinate. In the xy-plane, a parabola is a symmetric graph such that when two points have the same y-coordinate, these points are equidistant from the vertex, and the x-coordinate of the vertex is halfway between the x-coordinates of these two points. Therefore, for the graph of $y = fx$, the points with x-coordinates -9 and 3 are equidistant from the vertex, h, k, and h is halfway between -9 and 3. The value that is halfway between -9 and 3 is $\frac{-9+3}{2}$, or -3. Therefore, $h = -3$. The equation defining f can also be written in vertex form, $fx = a(x - h)^2 + k$. Substituting -3 for h in this equation yields $fx = a(x - -3)^2 + k$, or $fx = a(x + 3)^2 + k$. This equation is equivalent to $fx = ax^2 + 6x + 9 + k$, or $fx = ax^2 + 6ax + 9a + k$. Since $fx = ax^2 + 4x + c$, it follows that $6a = 4$ and $9a + k = c$. Dividing both sides of the equation $6a = 4$ by 6 yields $a = \frac{4}{6}$, or $a = \frac{2}{3}$. Since $\frac{2}{3} < 1$, it's not true that $a \geq 1$. Therefore, statement II isn't true. Substituting $\frac{2}{3}$ for a in the equation $9a + k = c$ yields $9\frac{2}{3} + k = c$, or $6 + k = c$. Subtracting 6 from both sides of this equation yields $k = c - 6$. If $k < 0$, then $c - 6 < 0$, or $c < 6$. Since c could be any value less than 6, it's not necessarily true that $c < 0$. Therefore, statement I isn't necessarily true. Thus, neither I nor II must be true.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

▼ 2.85 — D

Choice D is correct. For a function of the form $fx = a(b)^x$, where a and b are constants and $b < 1$, the value of $f(x)$ decreases by $100(1 - b)\%$ for every increase of x by 1. In the given function, $a = 56$ and $b = 0.19$. Therefore, for the given function, the value of $f(x)$ decreases by $100(1 - 0.19)\%$, or 81%, for every increase of x by 1. It's given that the value of $f(n)$ is $p\%$ less than the value of $f(n - 1)$. The increase from $x = n - 1$ to $x = n$ can be expressed as $n - (n - 1)$, which is equivalent to $n - n + 1$, or 1. Since there is an increase by 1 from $n - 1$ to n, it follows that the value of $f(n)$ is 81% less than the value of $f(n - 1)$. Thus, the value of p is 81.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

▼ 2.86 — 25/4, 6.25

The correct answer is $\frac{25}{4}$. The given equation can be rewritten in the form $fx = ax - h^2 + k$, where a, h, and k are constants. When $a > 0$, h is the value of x for which fx reaches its minimum. The given equation can be rewritten as $fx = 4x^2 - \frac{50}{4}x + 126$, which is equivalent to $fx = 4x^2 - \frac{50}{4}x + \frac{50}{8}^2 - \frac{50}{8}^2 + 126$. This equation can be rewritten as $fx = 4x - \frac{50}{8}^2 - \frac{50}{8}^2 + 126$, or $fx = 4x - \frac{50}{8}^2 - 4\frac{50}{8}^2 + 126$, which is equivalent to $fx = 4x - \frac{25}{4}^2 - \frac{121}{4}$. Therefore, $h = \frac{25}{4}$, so the value of x for which fx reaches its minimum is $\frac{25}{4}$. Note that 25/4 and 6.25 are examples of ways to enter a correct answer.